Day 12 — React Routing

Beginner notes and a complete multi-page React Router project

1. What is React Router?

React Router lets a React app show different components for different URL paths. Client-side navigation changes the displayed component without reloading the entire HTML page.

2. Install React Router

```
npm install react-router-dom
```

3. BrowserRouter

BrowserRouter uses the browser URL and History API to keep the UI synchronized with the current path. Wrap the app once near the root.

```
import { BrowserRouter } from "react-router-dom";

<BrowserRouter>
  <App />
</BrowserRouter>
```

4. Routes and Route

Routes selects the matching Route. Each Route maps a path to a React component through the element prop.

```
import { Routes, Route } from "react-router-dom";

<Routes>
  <Route path="/" element={<Home />} />
  <Route path="/about" element={<About />} />
  <Route path="*" element={<NotFound />} />
</Routes>
```

5. Link and NavLink

Link navigates without a full page reload. NavLink does the same and supports an active class when its route matches.

```
import { Link, NavLink } from "react-router-dom";

<Link to="/about">About</Link>
<NavLink to="/students">Students</NavLink>
```

6. Dynamic route parameters

A route such as /students/:studentId contains a dynamic segment. useParams reads that value from the URL.

```
<Route path="/students/:studentId" element={<StudentDetails />} />

import { useParams } from "react-router-dom";
const { studentId } = useParams();
```

7. Basic 404 page

Use path="*" as a catch-all route for unknown URLs.

```
<Route path="*" element={<NotFound />} />
```

Practical: Multi-page student app

• Home page

• About page

• Students list with sample students

• Dynamic student details page using useParams

• Contact page

• 404 Not Found page

Setup commands

```
npm create vite@latest day12-routing -- --template react
cd day12-routing
npm install
npm install react-router-dom
npm run dev
```

If using the supplied ZIP instead, extract it, open its folder in VS Code, and run npm install followed by npm run dev.

Project structure

```
src/
  main.jsx
  App.jsx
  App.css
  pages/
    Home.jsx
    About.jsx
    Students.jsx
    StudentDetails.jsx
    Contact.jsx
    NotFound.jsx
```

Testing checklist

■ Navigate between Home, About, Students, and Contact; the page should not fully reload.

■ Open a student's details; the URL should contain that student's ID.

■ Open /students/999; a Student not found message should appear.

■ Open /not-a-real-page; the 404 page should appear.

■ Use Return home to navigate back.

Note: Student information is hard-coded for practice. No backend or database is connected.